Sequence learning

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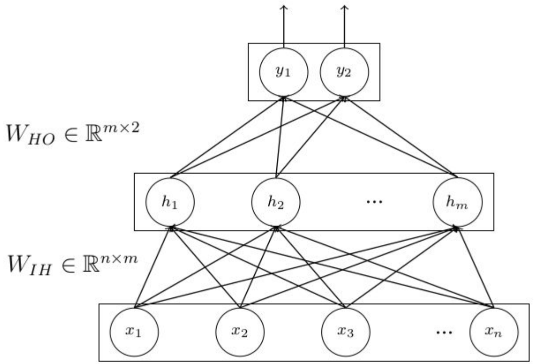
Sequence-learning and FFN

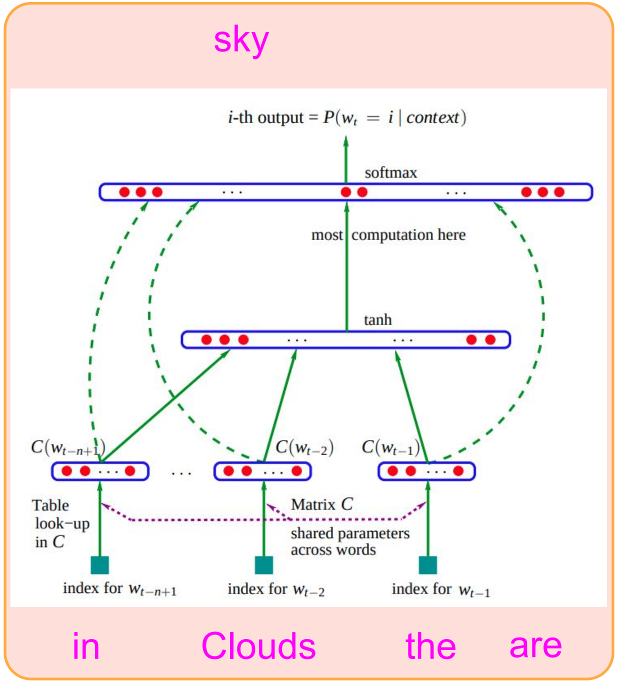
- Feed-Forward Network (FFN or MLP) is not designed to handle sequential or spatial data.
    - No sense or notion of sequence or location

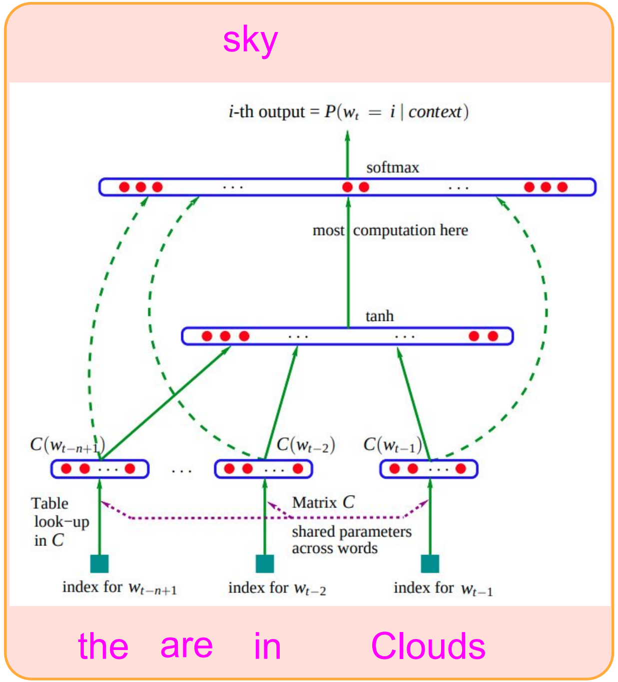


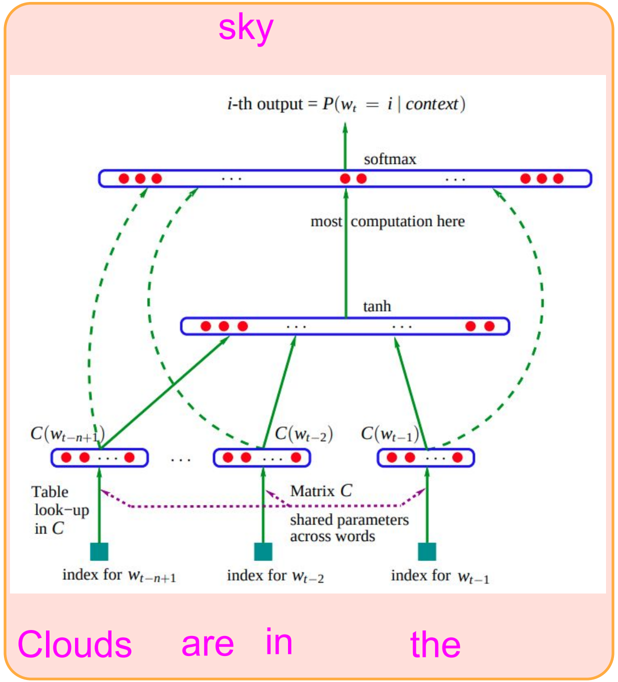
- If we want to generate numbers (*X*) in the sequence, it can't do it.
    - 3, 6, 9, 12, 15, 18, 21, 24, 27, 30, …*X*..

- For a FNN, there is no difference among the following input instances:
    - 3, 6, 9, 12, 15, 18, 21, 24, 27, 30
    - 30, 27, 24, 21, 18, 15, 12, 9, 6, 3
    - 3, 30, 24, 12, 6, 21, 15, 27, 9, 18

Sequence-learning and FFN





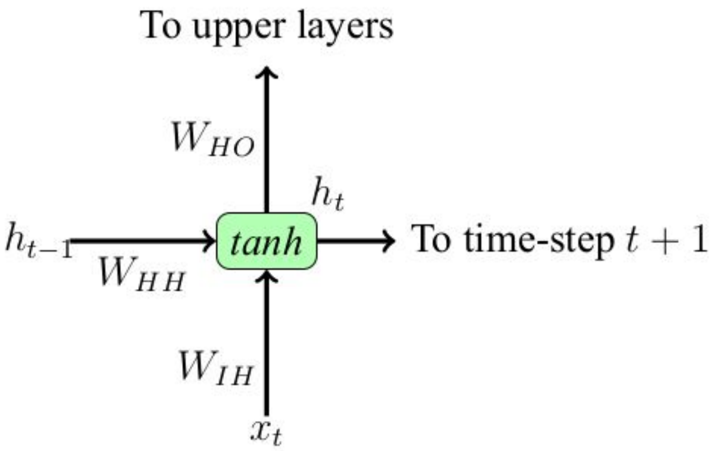


All these are essentially the same.

Recurrent Neural Network (RNN)

- A neural network with feedback connections
- Enable networks to do temporal/sequential processing
- Good at learning sequences
- Acts as memory unit

$$
\begin{aligned}
h_t &= tanh(W_{IH} \cdot x_t + W_{HH} \cdot h_{t-1} + b) \\
&= tanh([W_{IH}, W_{HH}] \cdot [x_t, h_{t-1}] + b)
\end{aligned}
$$



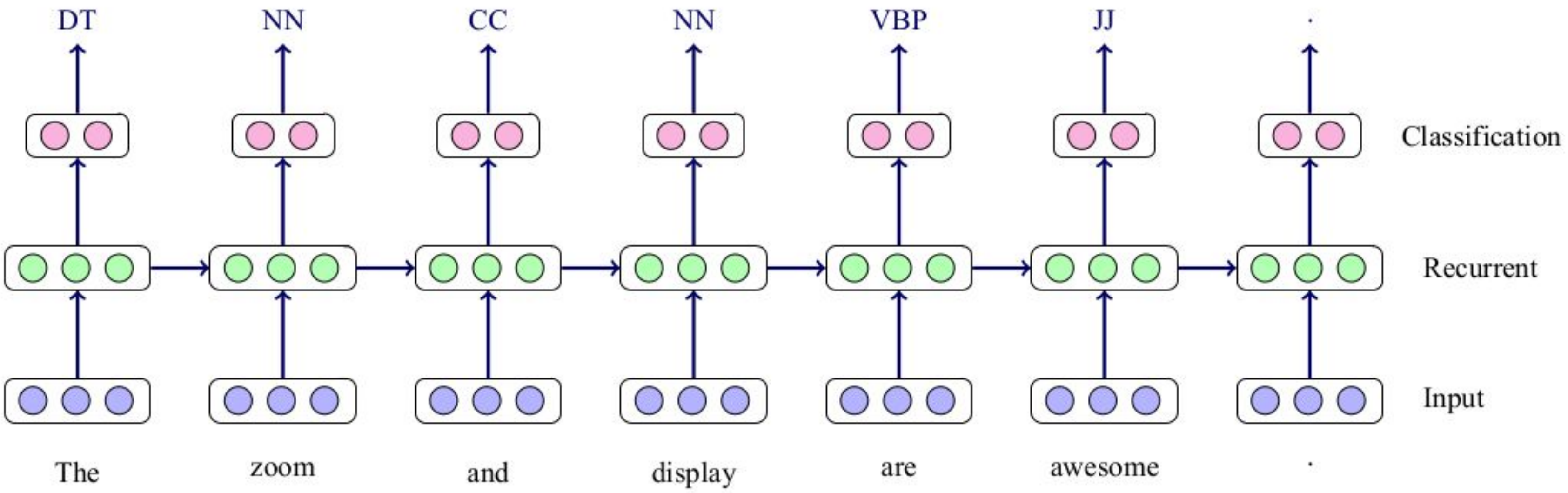
Usage

- Depends on the problems that we aim to solve.
- Typically good for sequence processings.
- Some sort of memorization is required.

RNN/LSTM/GRU for Sequence Labelling

Part-of-speech tagging:

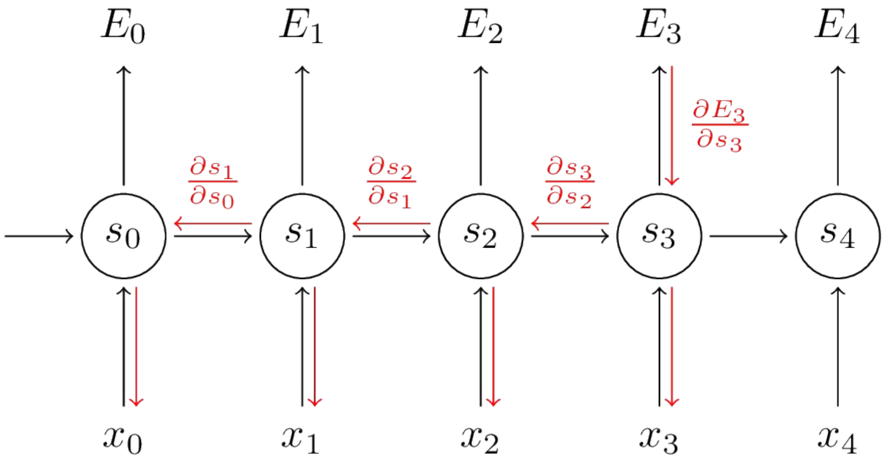
- Given a sentence X, tag each word its corresponding grammatical class.



Training of RNNs

How to train RNNs?

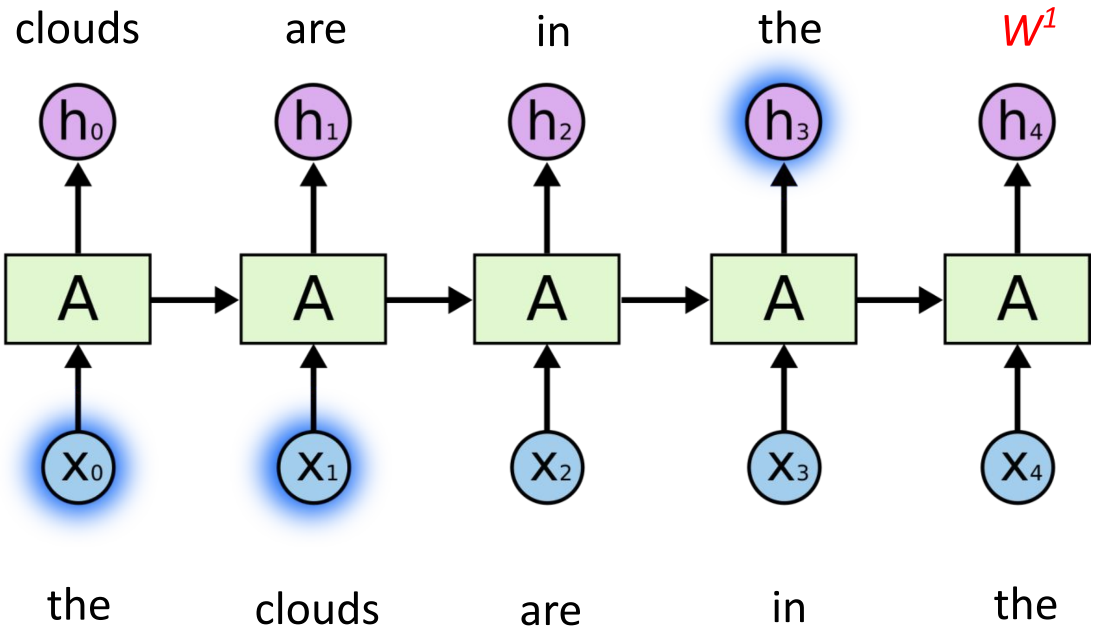
- Typical FFN
  - Backpropagation algorithm
- RNNs
  - A variant of backpropagation algorithm namely **Back-Propagation Through Time (BPTT)**.



Problems with RNNs

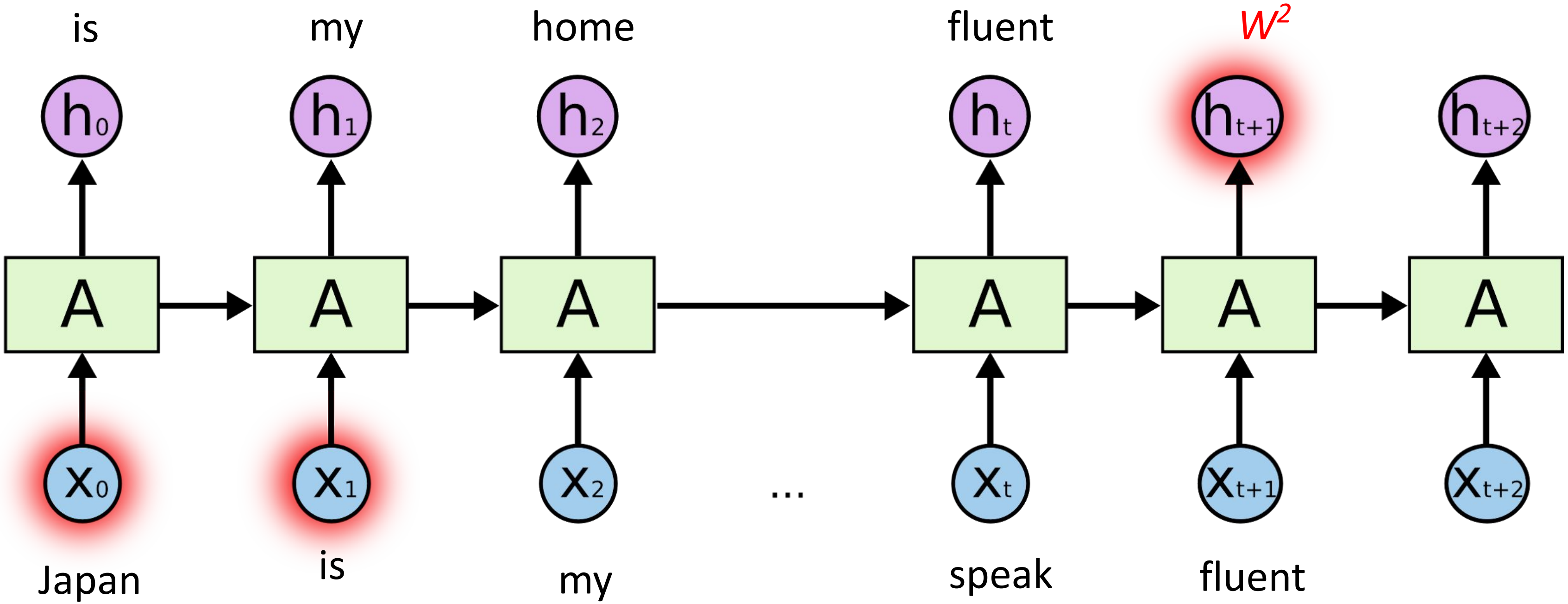
Language modelling: Example - 1

- "the clouds are in the *sky*"



Language modelling: Example - 2

- "Japan is my home country. I can speak fluent *Japanese*."



Vanishing/Exploding gradients

- Cue word for the prediction
  - Example 1: ***sky*** → ***clouds*** [3 units apart]
  - Example 2: ***Japanese*** → ***Japan*** [9 units apart]

- As the sequence length increases, it becomes hard for RNNs to learn “long-term dependencies.”
  - **Vanishing gradients:** If weights are small, gradient shrinks exponentially. Network stops learning.
  - **Exploding gradients:** If weights are large, gradient grows exponentially. Weights fluctuate and become unstable.

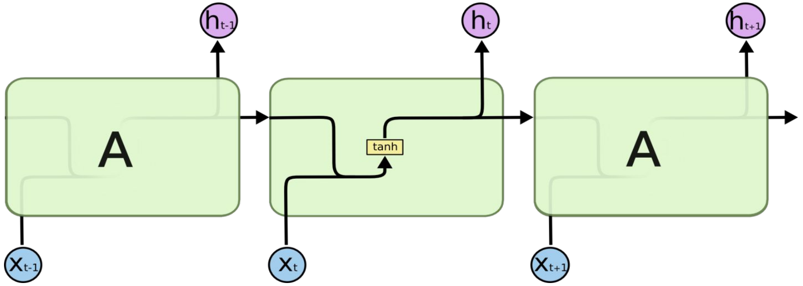
Long Short Term Memory (LSTM)

[Hochreiter & Schmidhuber (1997)]

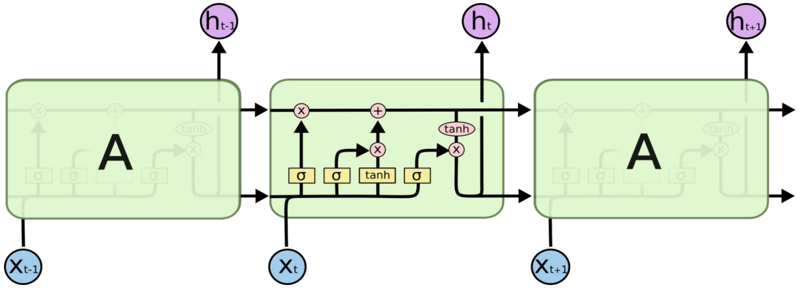
LSTM

- A variant of simple RNN (Vanilla RNN)
- Capable of learning long dependencies.
- Regulates information flow from recurrent units.

Vanilla RNN cell



LSTM cell



An LSTM cell

- Cell state c_t (blue arrow), hidden state h_t (green arrow) and input x_t (red arrow)
- Three gates
  - Forget (Red-dotted box)

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t])$$

$$c_t = f_t \otimes c_{t-1}$$

  - Input (Green-dotted box)

$$z_t = tanh(W_z \cdot [h_{t-1}, x_t])$$

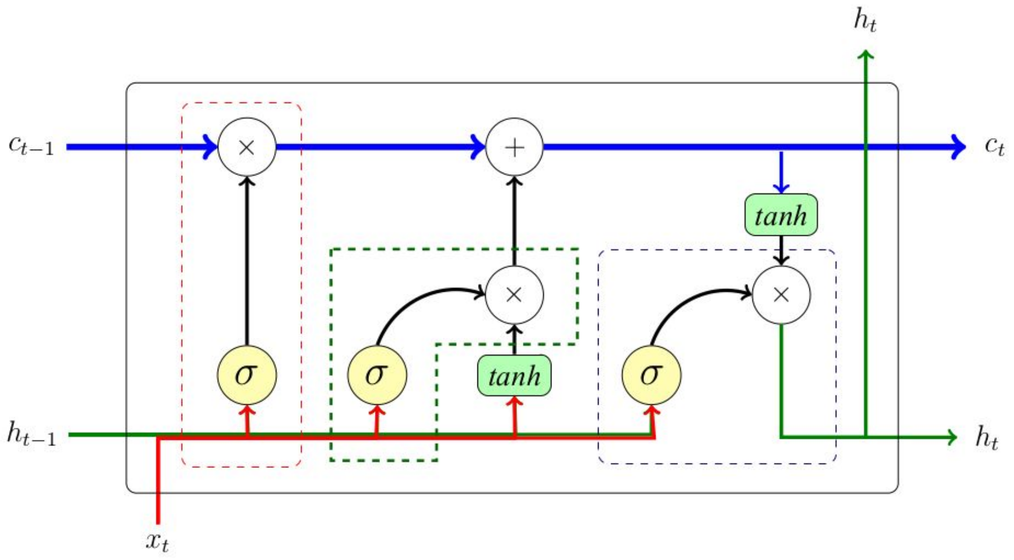
$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t])$$

$$c_t = c_t + (i_t \otimes z_t)$$

  - Output (Blue-dotted box)

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t])$$

$$h_t = o_t \otimes tanh(c_t)$$

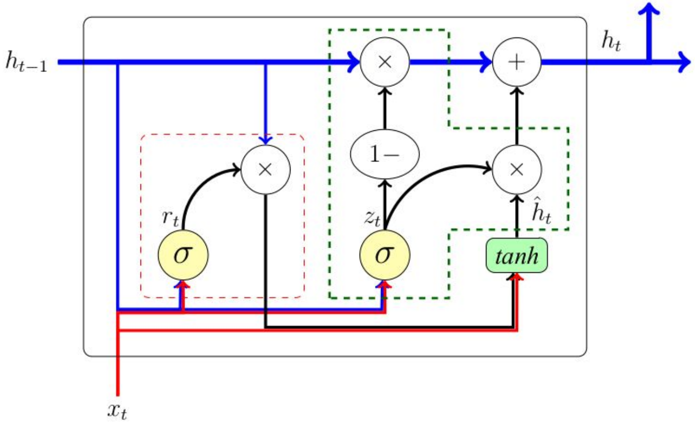


Gated Recurrent Units (GRU)

Kyunghyun Cho, Bart van Merrienboer, Caglar Gulcehre, Dzmitry Bahdanau, Fethi Bougares, Holger Schwenk, Yoshua Bengio (2014)

Gated Recurrent Unit (GRU) [Cho et al. (2014)]

- A variant of simple RNN (Vanilla RNN)
- Similar to LSTM
  - Whatever LSTM can do GRU can also do.
- Differences
  - Cell state and hidden are merged together
  - Two gates
    - Reset gate - similar to forget
    - Update gate - similar to input gate
  - No output gate
  - Cell/Hidden state is completely exposed to subsequent units.
- GRU needs fewer parameters to learn and is relatively efficient *w.r.t.* computation.



$$
\begin{aligned}
z_t &= \sigma(W_z \cdot [h_{t-1}, x_t]) \\
r_t &= \sigma(W_r \cdot [h_{t-1}, x_t]) \\
\hat{h}_t &= tanh(W \cdot [r_t \otimes h_{t-1}, x_t]) \\
h_t &= (1 - z_t) \otimes h_{t-1} + z_t \otimes \hat{h}_t
\end{aligned}
$$

LSTM layer in TensorFlow vs PyTorch

```
tf.keras.layers.LSTM(
    units,
    activation='tanh',
    recurrent_activation='sigmoid',
    use_bias=True,
    dropout=0.0,
    recurrent_dropout=0.0,
    return_sequences=False,
    return_state=False,
    ..
    ..
    ..
    **kwargs
)
```

```
torch.nn.LSTM(
    self,
    input_size,
    hidden_size,
    num_layers=1,
    bias=True,
    batch_first=False,
    dropout=0.0,
    bidirectional=False,
    proj_size=0,
    device=None,
    dtype=None
)
```

https://pytorch.org/docs/stable/generated/torch.nn.LSTM.html

*Many more options are available
https://www.tensorflow.org/api_docs/python/tf/keras/layers/LSTM

Applications of RNNs in NLP

Tasks

- RNNs can be applied for sequence to sequence (seq-2-seq) problems

$$\text{RNN}(x_1, x_2, \ldots, x_n) \qquad \Rightarrow \qquad y_1, y_2, \ldots, y_m$$

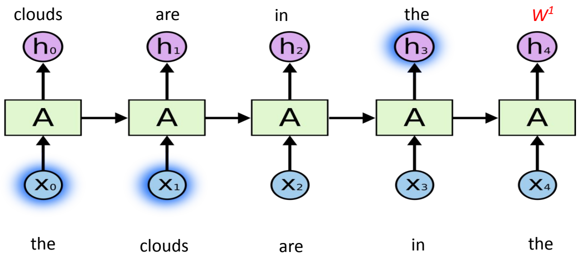
where n >= 1 and m >= 1

- In general, we can use RNNs for
    - Sequence Transformation (Generation) ⇒ n >= 1 and m >= 1
    - Sequence Labelling ⇒ n >= 1 and m >= 1 and $\boldsymbol{n}$ **==** $\boldsymbol{m}$
    - Classification ⇒ n >= 1 and m == 1

Sequence Labelling Tasks

- By definition, for each input timestep, we have one output.
  - At each time step, it's a classification task only. :)
- Tasks:
  - Language Modelling
  - PoS Tagging
  - Information extraction tasks
    - Named-Entity Recognition
    - Entity and Relation Extraction
    - Answer Extraction
    - …

Language Modelling



- Very simple and a trivial task.

- Outcome: The model learns the linguistic properties of a language.

- However, needs huge amount of data
- But, the advantage is we don't need explicit* training data
    - We can easily compute training data from the raw text, which is abundant.

[John ate an apple .] → [ate an apple . <eos>]

Masked Language Modelling

- Similar to the standard LM task with minor difference.

- Outcome: The model learns the linguistic properties of a language.

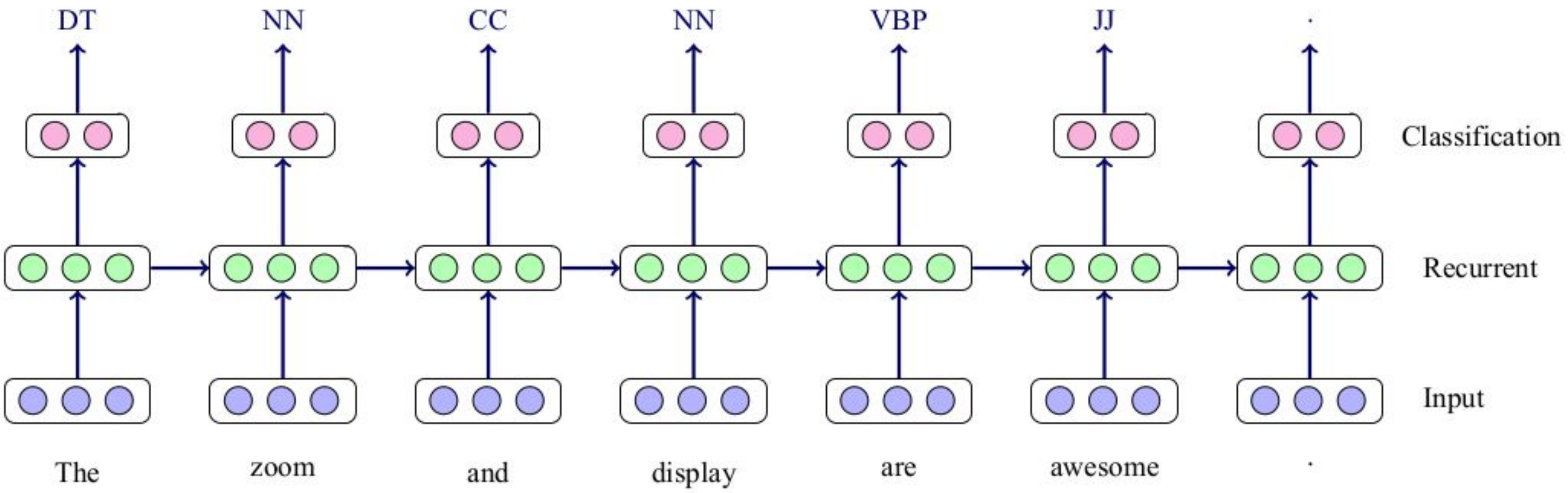
- However, needs huge amount of data
- But, the advantage is we don’t need explicit* training data
    - We can easily compute training data from the raw text, which is abundant.

[John <masked> an apple .] → [John ate an apple .]

- Minor difference in the objective:
    - LM: Predicts the next word in sequence.
    - MLM: Predicts the masked word(s) to complete the sentence.

Part-of-Speech Tagging

- Given a sentence X, tag each word its corresponding grammatical class.



Information Extraction - Named-Entity Recognition

- Filtering/extracting information (names) from the input.
  - Output must be present in the input.

India	born	Sundar	Pichai	is	the	CEO	of	Google	and	its	parent	company	Alphabet	.

Find named-entities or proper nouns.

Y	N	**Y**	**Y**	N	N	N	N	**Y**	N	N	N	N	**Y**	N

Not enough! what kind?

Location	N	Person	Person	N	N	N	N	Organization	N	N	N	N	Organization	N

Information Extraction - Named-Entity Recognition

- What are possible types of named-entities?
  - Person
  - Location
  - Organization
  - Date and Time
  - Some other types … (we need to define it if we want it.)

Information Extraction - Named-Entity Recognition

- Modeling
  - In previous example

$$x_1, x_1, \ldots, x_n \quad \Rightarrow \quad y_1, y_1, \ldots, y_n \quad \text{where, } y_i \in \{\text{Per, Loc, Org}\}$$

India	born	Sundar	Pichai	is	the	CEO	of	Google	and	its	parent	company	Alphabet	.

Loc	N	Per	Per	N	N	N	N	Org	N	N	N	N	Org	N

How many organizations are there?

How many persons are there?

- We need to tell the model that
  - Sundar and Pichai are the same person
  - Google and Alphabet are different organizations

*Similar encoding schemes: IOB, IOB2

Begin-Intermediate-Other (BIO*) encoding

- For each label (i.e., Per, Loc, Org, etc.), we need two sub-labels:
  - B-Per, B-Loc, B-Org
  - I-Per, I-Loc, I-Org
  - O

India	born	Sundar	Pichai	is	the	CEO	of	Google	and	its	parent	company	Alphabet	.
B-Loc	O	B-Per	I-Per	O	O	O	O	B-Org	O	O	O	O	B-Org	O

Good for identifying multi-word expressions

*Similar encoding schemes: IOB, IOB2

Begin-Intermediate-Other (BIO*) encoding

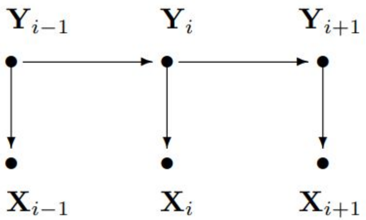
- For each label (i.e., Per, Loc, Org, etc.), we need two sub-labels:
  - B-Per, B-Loc, B-Org
  - I-Per, I-Loc, I-Org
  - O

I	saw	Sachin	Virat	and	Rahul	Dravid	at	Eden	Garden	.
O	O	B-Per	B-Per	O	B-Per	I-Per	O	B-Loc	I-Loc	O

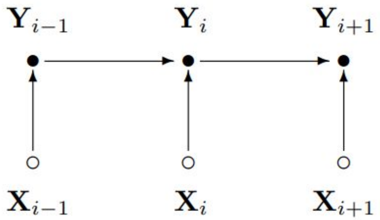
Good for distinguishing two consecutive distinct entities of same type.

Classical Sequence Learners

- Hidden Markov Model (HMM)
    - Generative

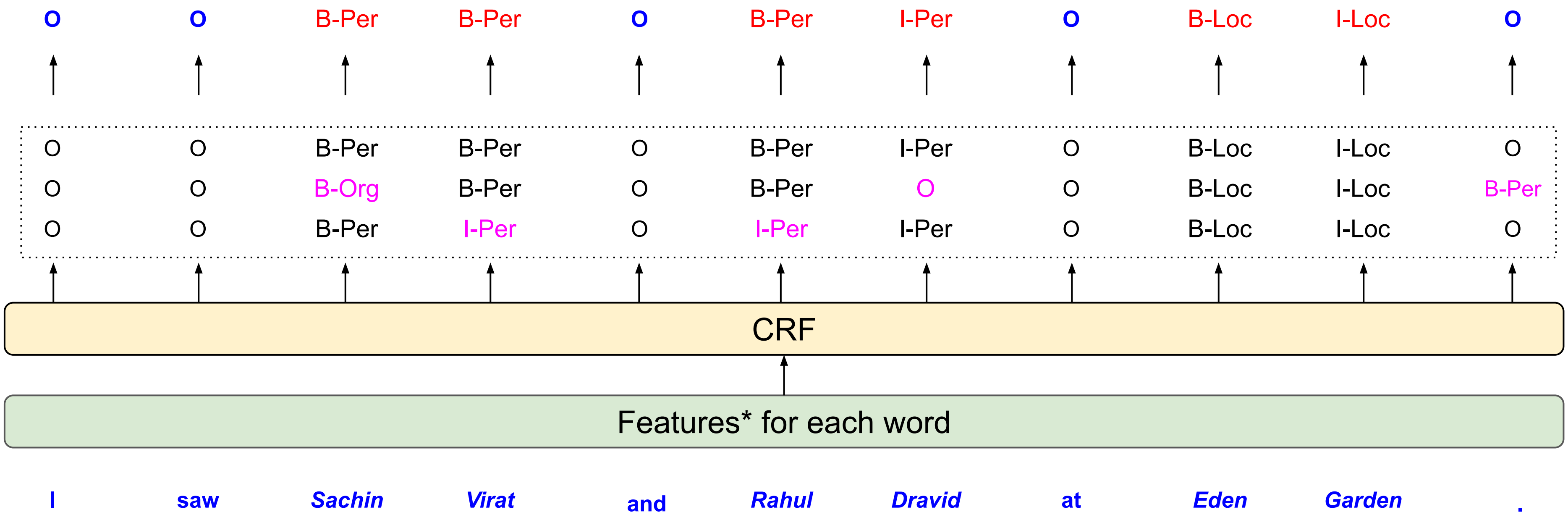


- Maximum Entropy Markov Model (MEMM)
    - Discriminative
    - MEMMs are trained locally to maximize the probability of each individual label



- Conditional Random Field (CRF)
    - Similar to MEMM but trained globally to maximize the probability of the overall sequence.

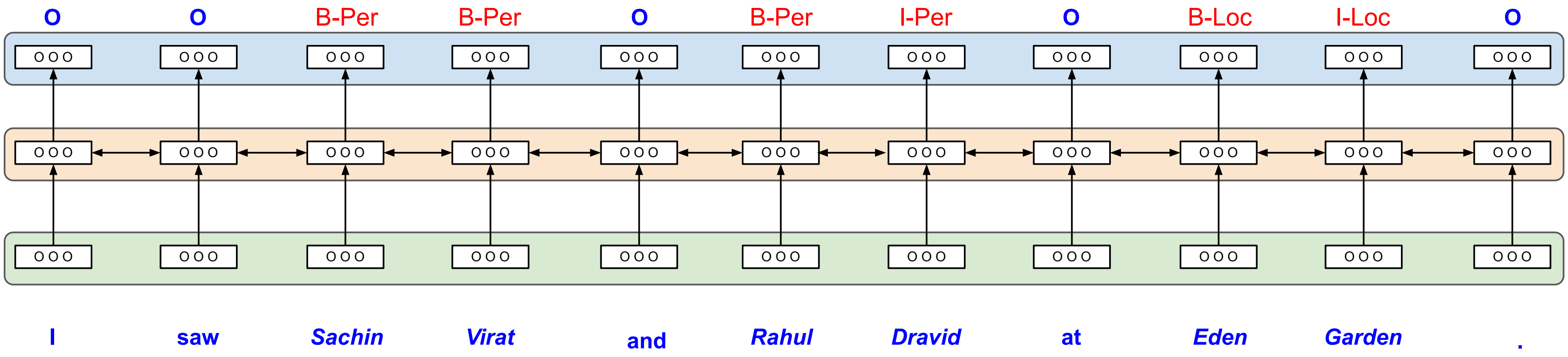
Classical Sequence Learners



– Gazetteers (common names, e.g. kumar, md)
– Word structure (Capitalization, etc.)
– PoS tags (Nouns)

RNN with BIO encoding

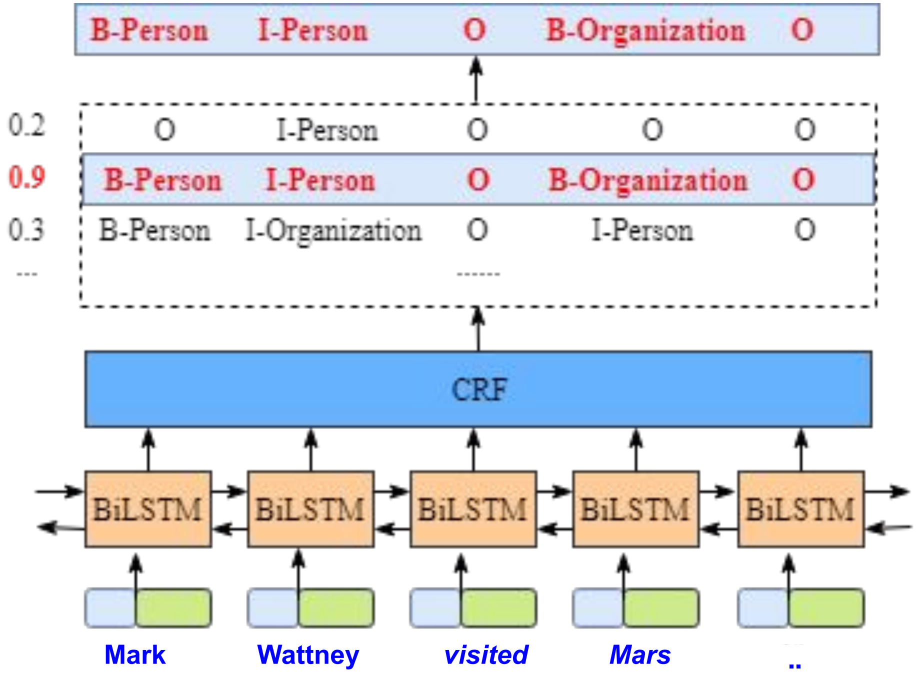
- Three layers
  - Embedding layer $\mathbf{S}_{WxD}$ W = no-of-words; D = dimension of each word
  - Bi-LSTM layer $\mathbf{L}_{WxH}$ H = Recurrent dimension
  - Softmax layer $\mathbf{P}_{WxC}$ C = 2*|labels| + 1



https://aclanthology.org/N16-1030.pdf

BiLSTM-CRF [Lample et al., 2016]

- Classically, CRF works remarkably well.
    - however, it needs features.
- Stacked
    - CRF on top a BiLSTM layer
- Utilized both word-level and char-level embeddings



https://createmomo.github.io/2017/09/12/CRF_Layer_on_the_Top_of_BiLSTM_1/

Information Extraction - Answer extraction

- Given a passage/context and a question,
  - Extract the answer from the passage OR
  - Identify the span where the answer is located.

Context: India born Sundar Pichai is the CEO of Google and its parent company Alphabet……
Question: Who is Sundar Pichai?

Answer Generation

Answer Extraction

Answer: CEO of Google and Alphabet.

Why should I trust you? Show me the proof.

India born Sundar Pichai is the CEO of Google and its parent company Alphabet ……

Span Identification

- We can model the tasks in one of following ways:
  - Sequence labelling
    - BIO encoding → Three classes *B*, *I*, and *O*
    - Boundary detection – Identify the start and end indexes of the answer → Three classes start, end, and none
      - It will be a multilabel classification problem to handle the span of a single token.
  - Generation
    - Generate the start and end indexes

Context: India born Sundar Pichai is the CEO of Google and its parent company Alphabet……
Question: Who is Sundar Pichai?

India_O born_O Sundar_B Pichai_I is_I the_I CEO_I of_I Google_I and_I its_I parent_I company_I Alphabet_I …

India_none born_none Sundar_start Pichai_none is_none the_none CEO_none of_none Google_none and_none its_none parent_none company_none Alphabet_end …

<start-token-index, end-token-index> ⇒ <2, 13>

Other Information Extraction Tasks

- Event Extraction
    - ***{Nepal earthquake}**$_{event}$ kills at least 128, toll could rise, officials.*
- Aspect-term Extraction
    - ***{Battery life}**$_{aspect\text{-}term}$ is awesome but **{camera}**$_{aspect\text{-}term}$ is dreadful.*
- Claim-span identification
    - *Whether made on purpose or not **{#coronavirus was used by the #CCP as a bio weapon}**$_{claim}$, not only to kill people but to encourage racism among their citizens against foreigners. Especially black people, **{CCP is kicking out black people from hotels even if they dont have covid}**$_{claim}$.*
- Hateful phrases (span) identification
    - *@user **This {trashy wh*** gold digger}**$_{Span}$ at again trying be something that makes her think shes good ughhh not happening **{h** bag pos}**$_{span}$*
- Privacy-related entity extraction
    - *When **{you}**source-direct **{visit the site}**medium, **{we}**target-direct also collect **{web site usage information ,the type and version of browser and operating system}**data-compulsory **{you}**source-direct use, **{if you arrived at trainchi-nese.com via a link from another website , the URL of the linking page}**data-compulsory. **{We}**target-direct use this information to **{ensure our site is compatible with the browsers used by most of our visitors}**reason and to **{improve the customer experience}**reason.*